Rhemac10 MERFISH Pipeline Overview

1. Merge all region cell_by_gene.csv's into one csv and all region cell_metadata.csv's of slide5 into one csv for slide5
2. Create adata python object from all_regions_cell_by_gene.csv & all_regions_cell_metadata.csv
3. Calculate statistics for adada object and create plots
4. Determine thresholds for transcripts per cell and volume per cell (filter adata)
5. Perform standard scanpy analysis downstream (normalize, scale, PCA, UMAP, DGE) & create spatial scatter plots with squidpy
6. Create transcripts per cell histograms
7. Detect ODCs in MERSCOPE Vizualizer. Export csv's for each region - identifies and stores list of cells that express genes of interest based on gene expression threshold

Create adata object

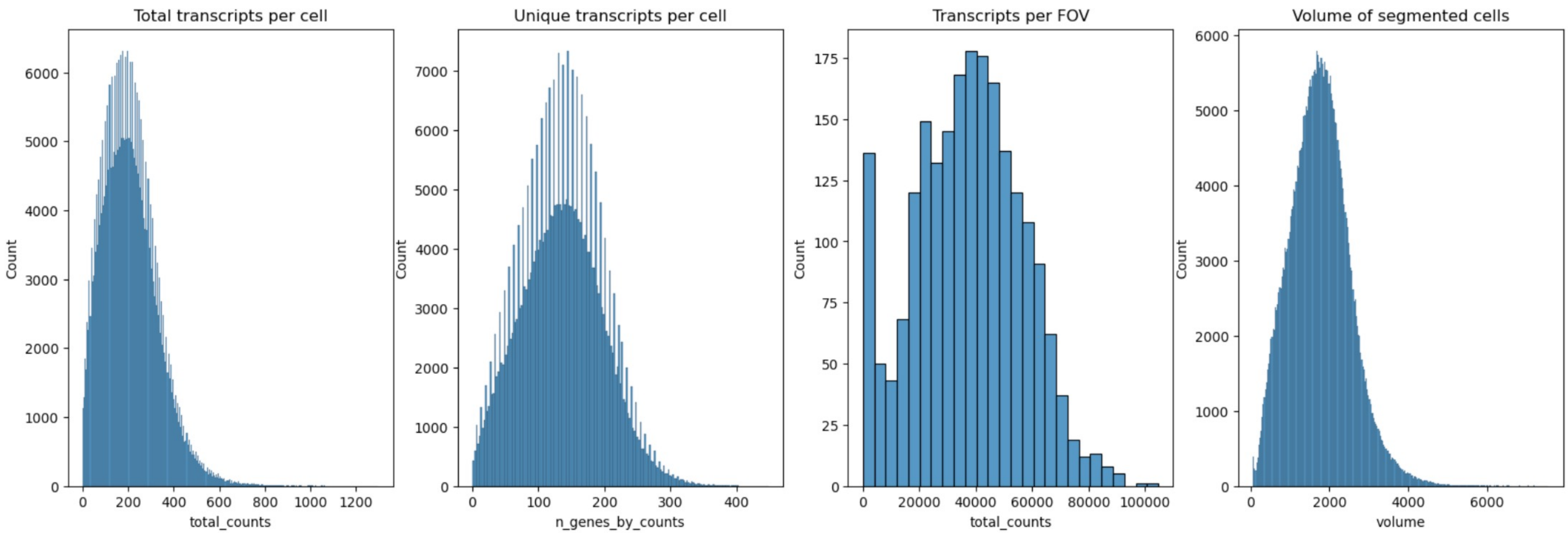
- Create python object called adata from merged cell by gene csv (counts matrix) and merged cell matadata csv (xyz coordinates, transcripts per cell, volumne per cell, etc)
- adata has 378,166 cells (rows) and 881 genes (columns) (from gene panel)

```
AnnData object with n_obs × n_vars = 378166 × 881
    obs: 'fov', 'volume', 'min_x', 'min_y', 'max_x', 'max_y', 'anisotropy', 'transcript_count', 'perimeter_area_ratio', 'solidity', 'MEF2C
_raw', 'MEF2C_high_pass', 'PolyT_raw', 'PolyT_high_pass', 'SLC17A7_raw', 'SLC17A7_high_pass', 'APOE_raw', 'APOE_high_pass', 'DAPI_raw', 'D
API_high_pass'
    uns: 'spatial'
    obsm: 'blank_genes', 'spatial'
```

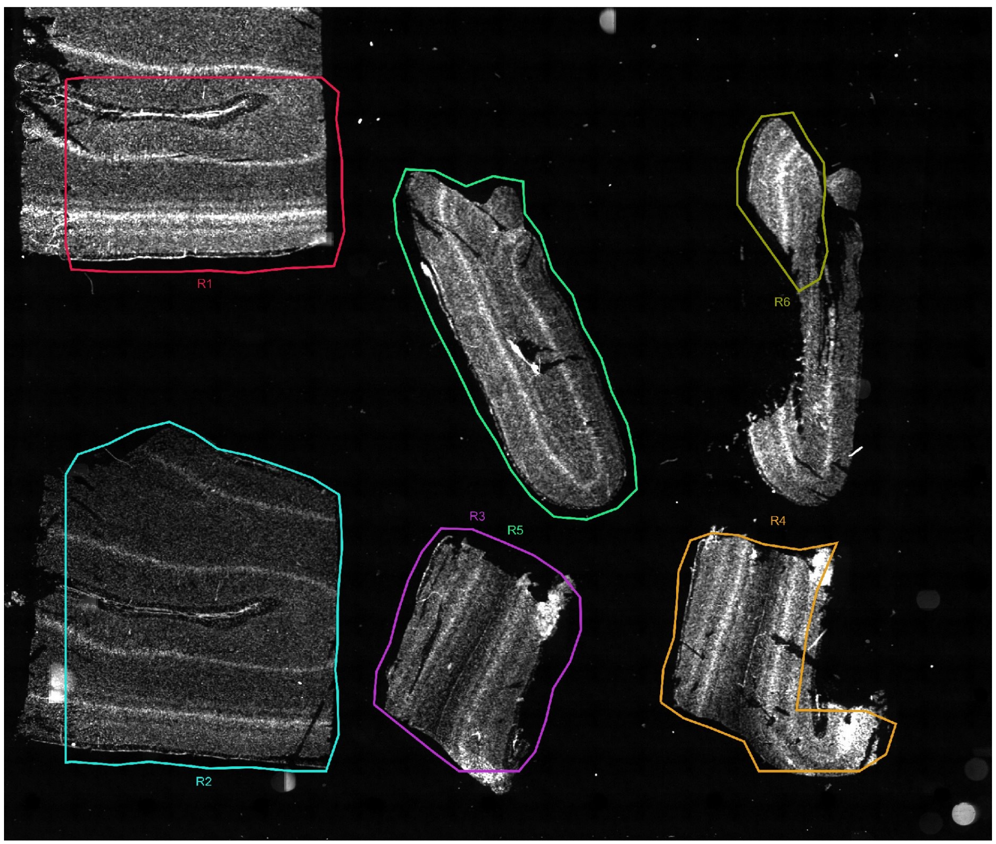
Adata object stats

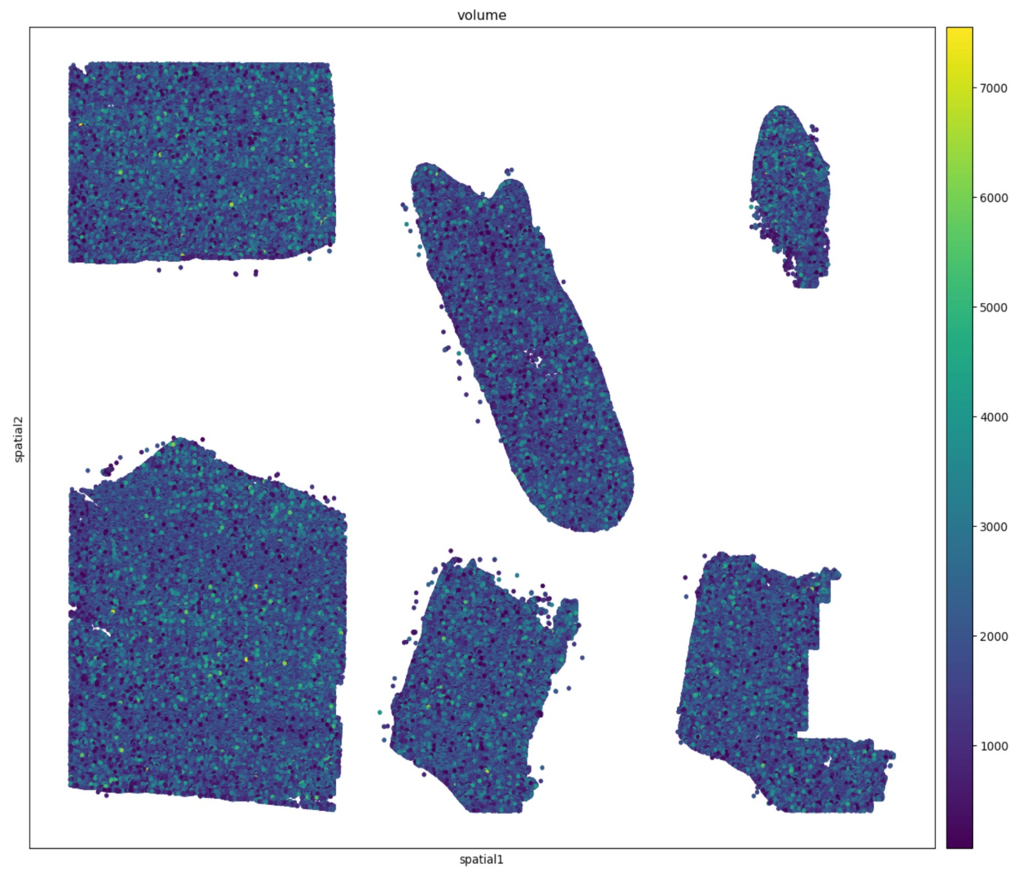
```
Fraction of non-empty cells: 100.00%
Fraction of empty cells: 0.00%
Number of cells: 378166
Total transcripts: 80307167
Median transcript/cell: 200.0
Median genes/cell: [[ 31.  25.   7. ... 215. 199. 216.]]

mean transcript count per cell: 212.35956431831525
max transcript count per cell: 1302
2% quantile of transcript count: 23.0
98% quantile of transcript count: 506.0
```

QC Histograms

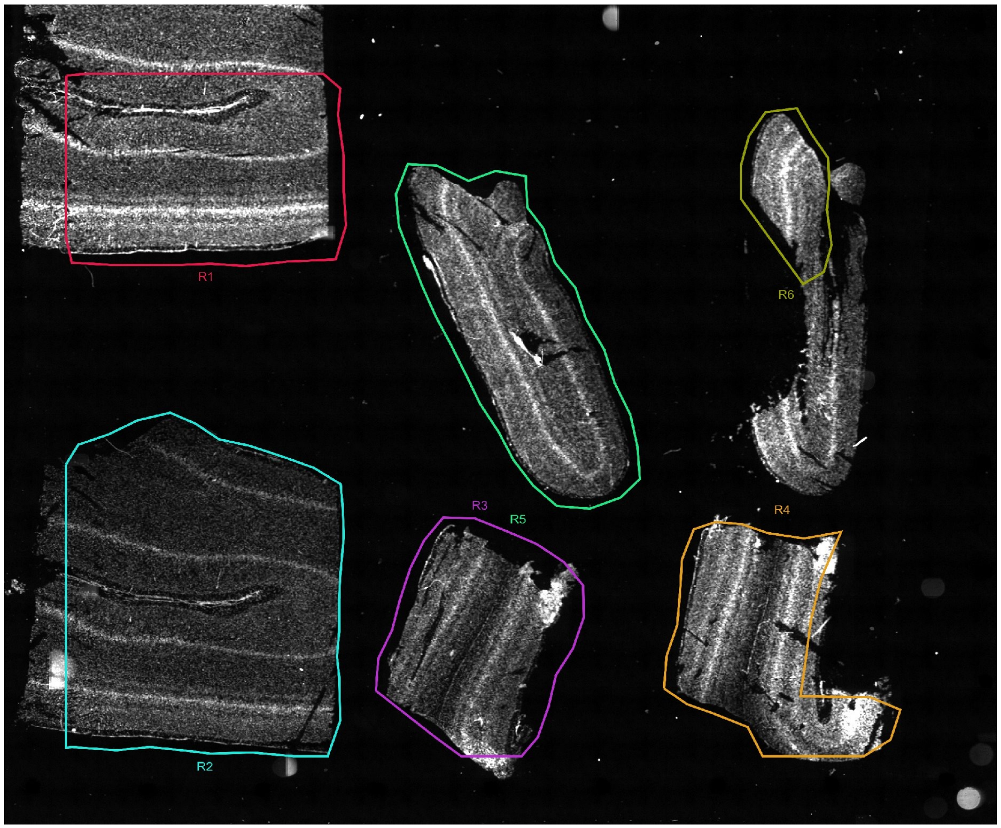


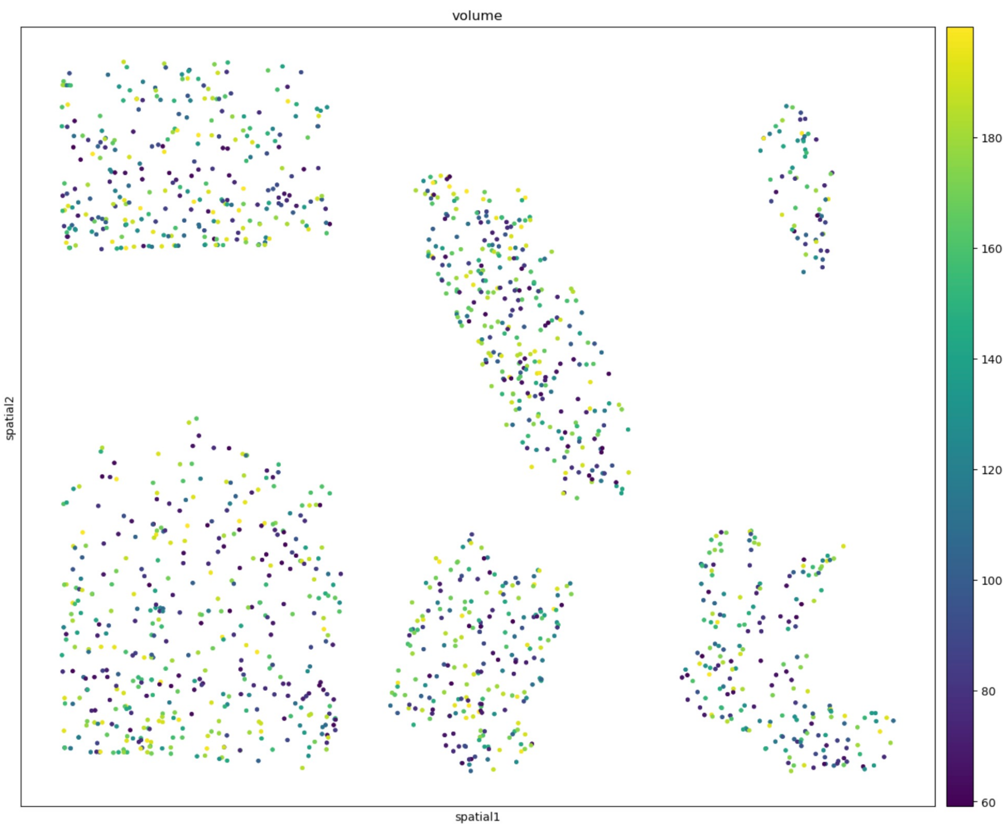
Spatial distribution of volumes across tissue



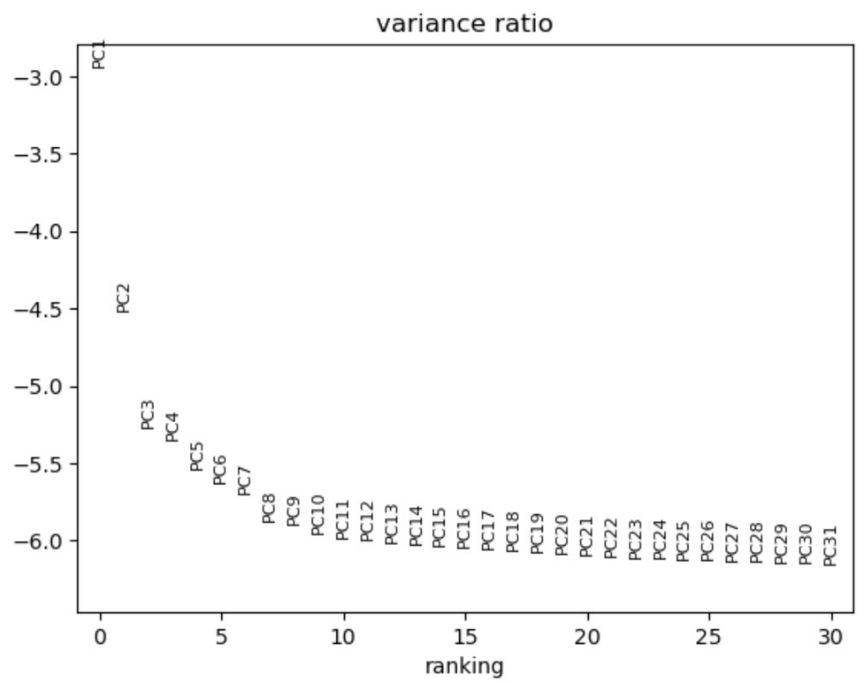


Spatial distribution of volumes less than 200 um3 across the tissue



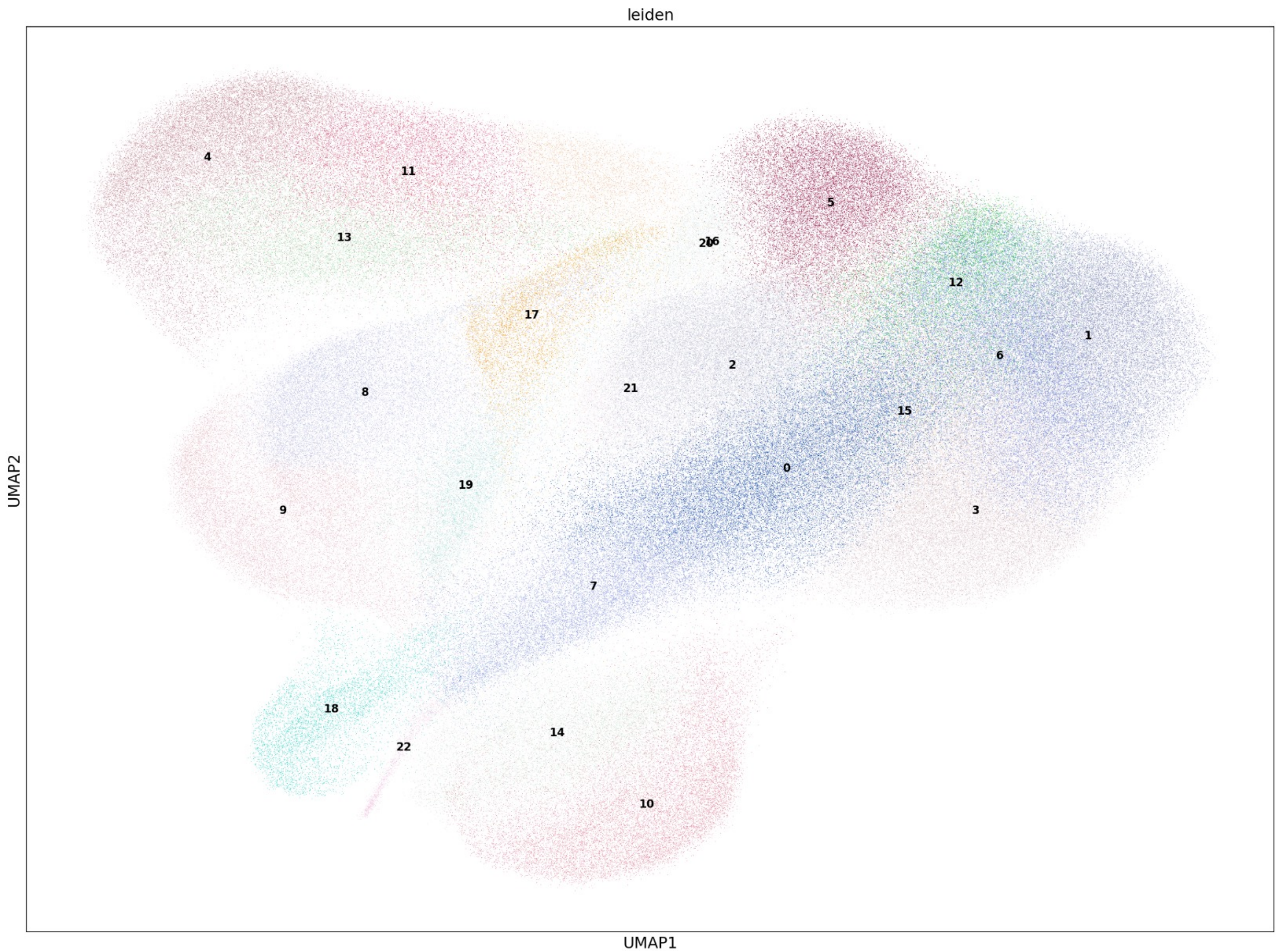


Run PCA

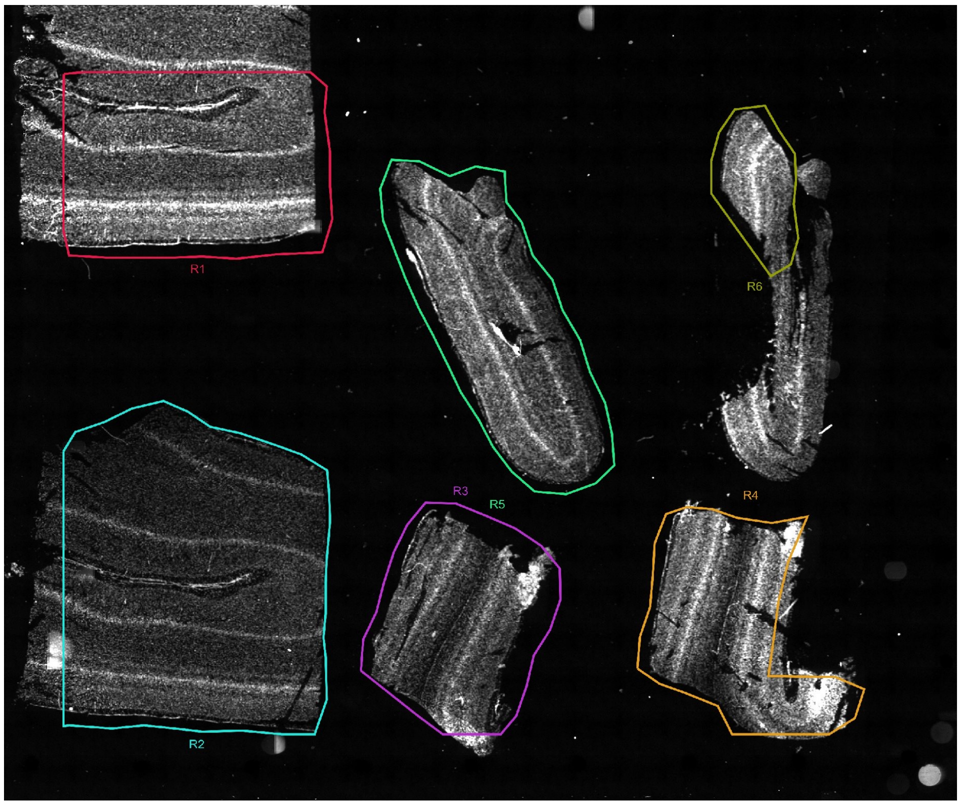

variance ratio
−3.0
−3.5
−4.0
−4.5
−5.0
−5.5
−6.0
PC1
PC2
PC3
PC4
PC5
PC6
PC7
PC8
PC9
PC10
PC11
PC12
PC13
PC14
PC15
PC16
PC17
PC18
PC19
PC20
PC21
PC22
PC23
PC24
PC25
PC26
PC27
PC28
PC29
PC30
PC31
0
5
10
15
20
25
30
ranking

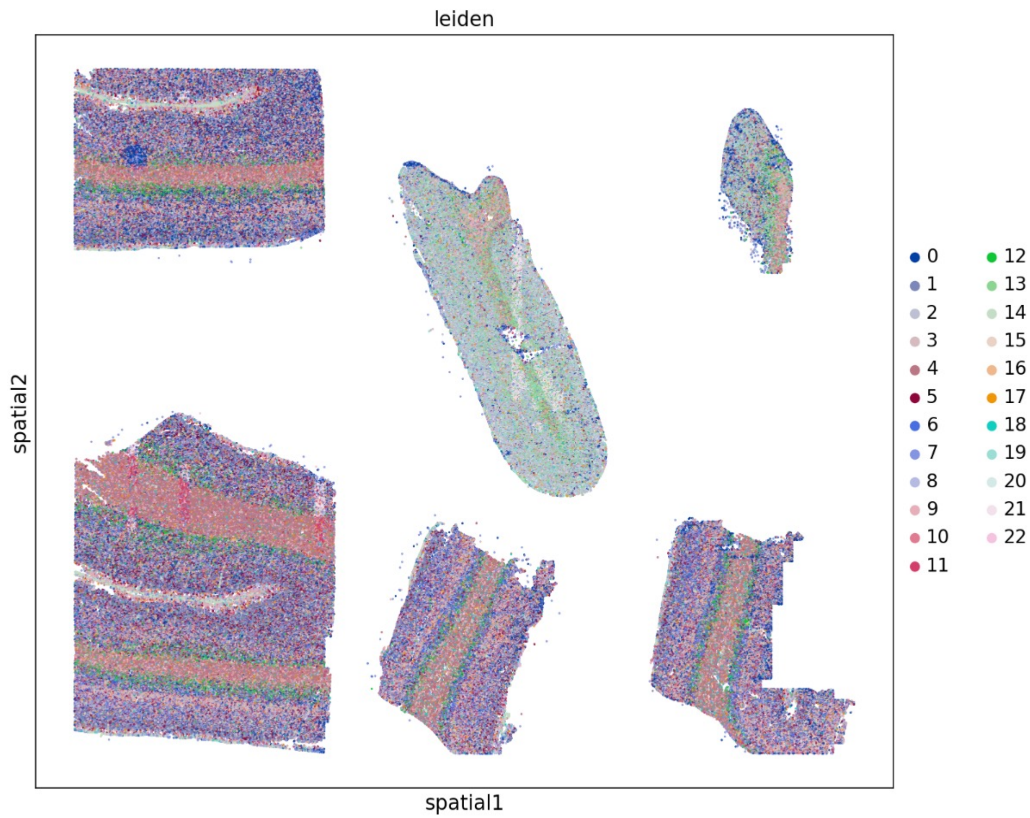
UMAP

Resolution = 1.5
PCs = 1-15
23 clusters total

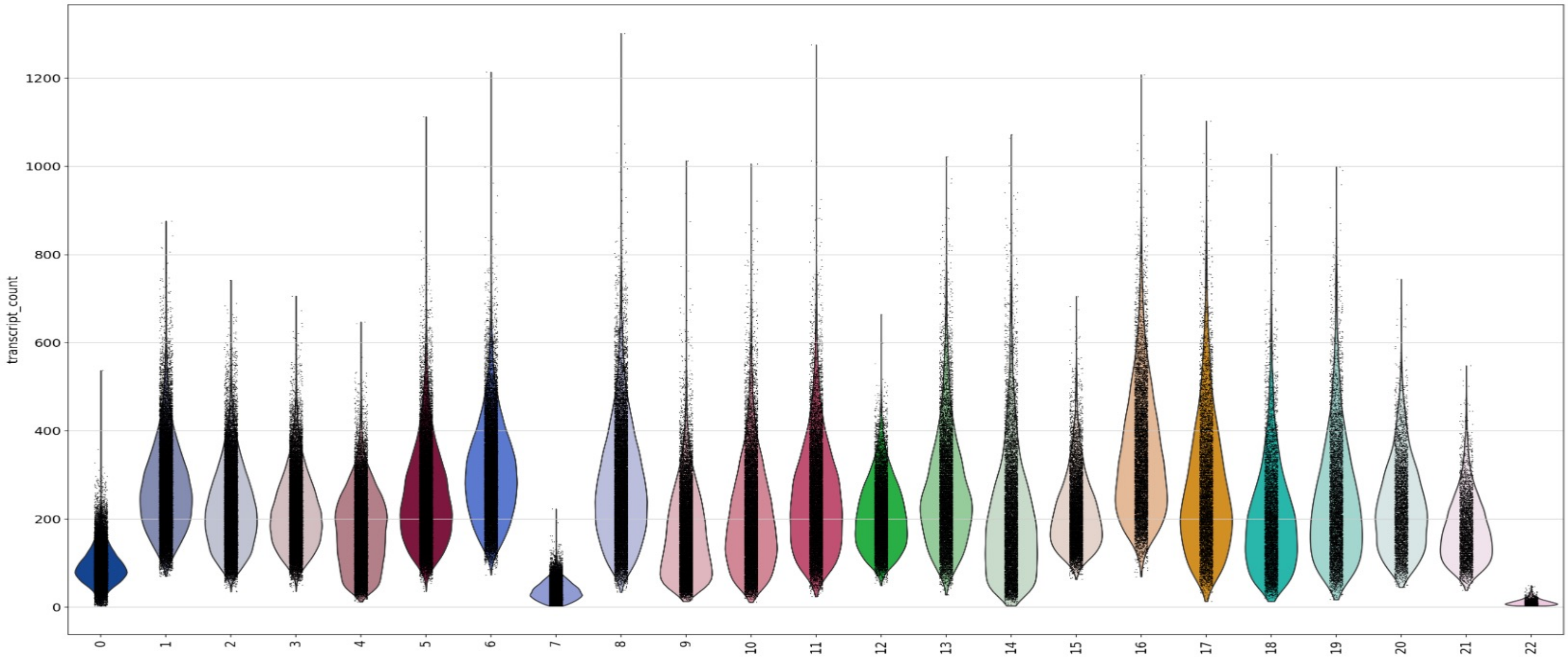


UMAP spatial clusters on tissue



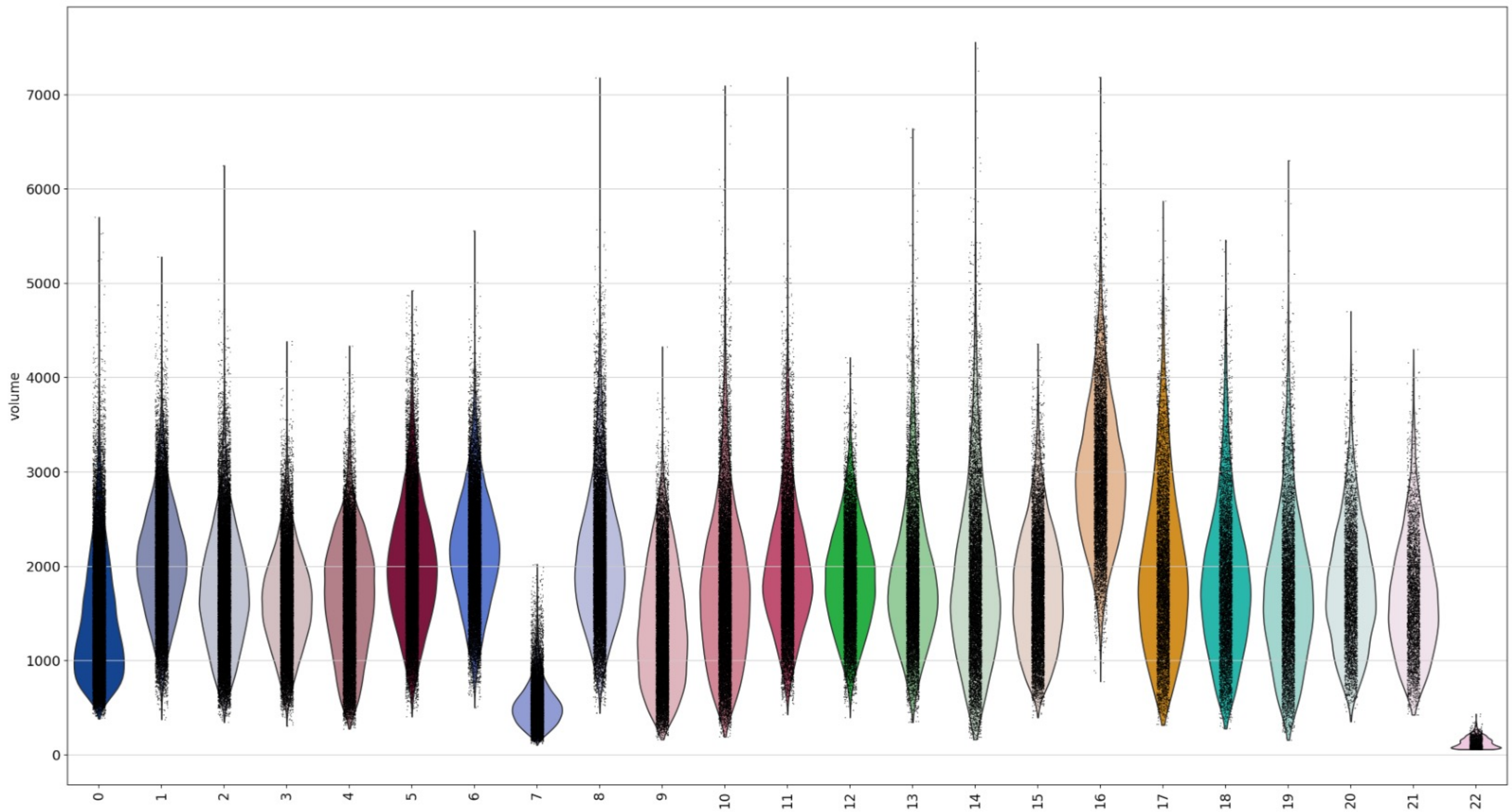


Transcript count by cluster violin plot



Percentage of cells with less than 25 counts in leiden groups 0: 2.070313656961763
Percentage of cells with less than 25 counts in leiden groups 4: 0.2029797426216864
Percentage of cells with less than 25 counts in leiden groups 7: 28.80228605212593
Percentage of cells with less than 25 counts in leiden groups 14: 1.1962782454585734
Percentage of cells with less than 25 counts in leiden groups 22: 95.21707137601177

Volume by cluster violin plot



Percentage of cells with volume less than 200 in leiden groups 7: 1.8377100064048875

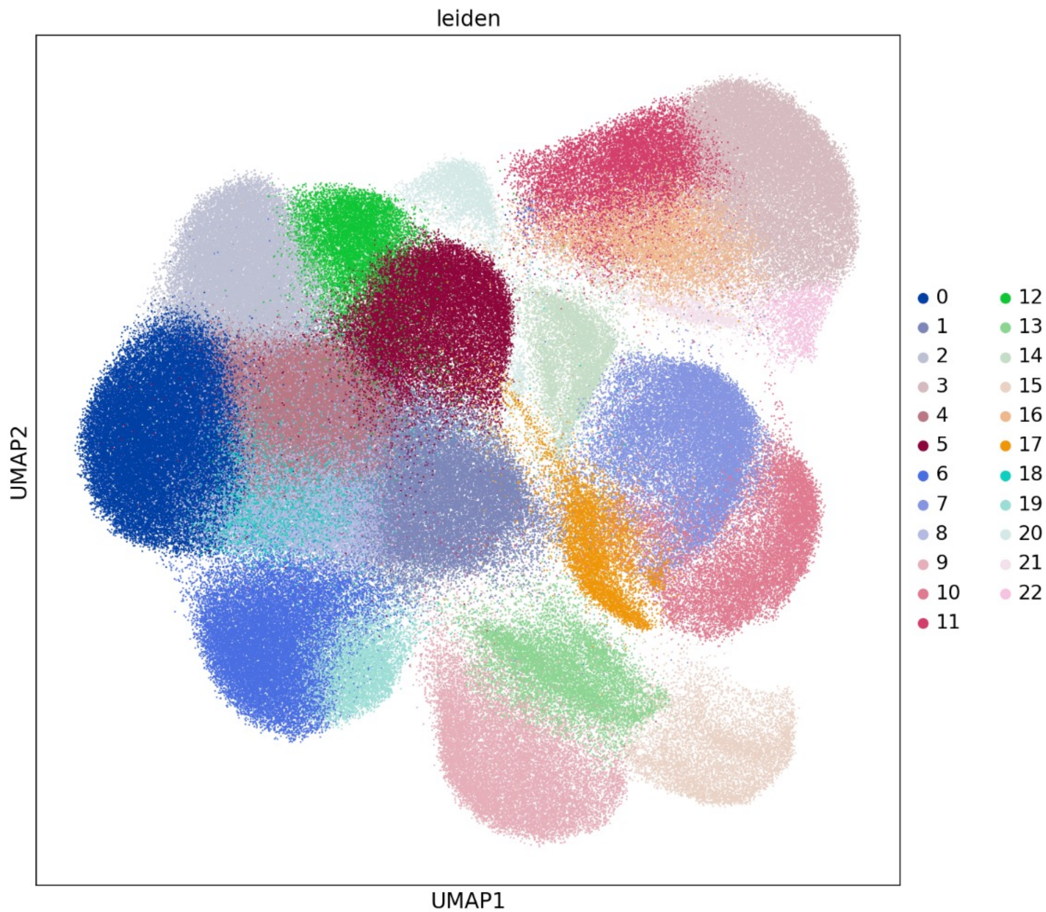
Percentage of cells with volume less than 200 in leiden groups 22: 83.9587932303164

Create adata_filter object

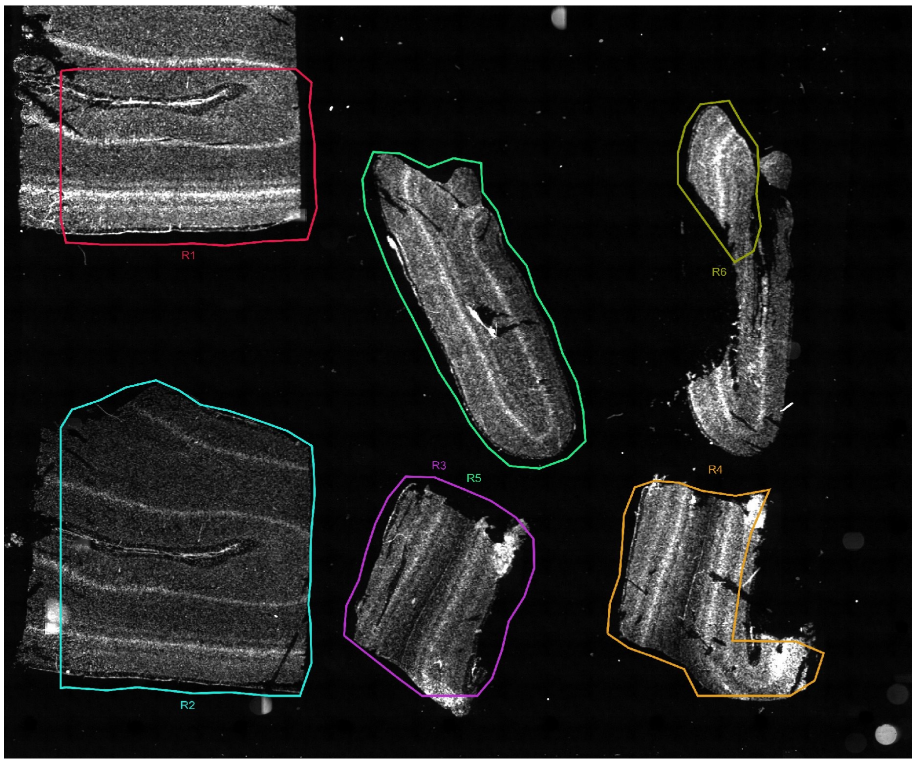
- Make new adata_filter object with cells that have min of 25 mRNA transcripts, min vol of 200, and max vol of 3000
- 29,706 cells filtered out
- Total cells adata_filter = 348,460

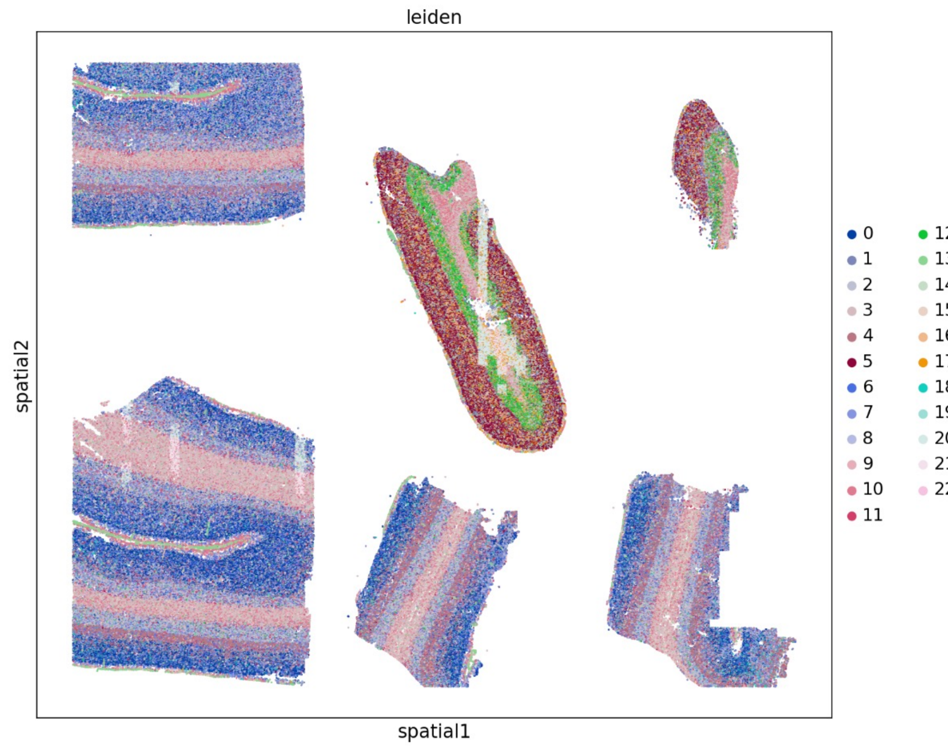
UMAP adata_filter

Resolution = 1.5
PCs = 1-15
23 clusters total

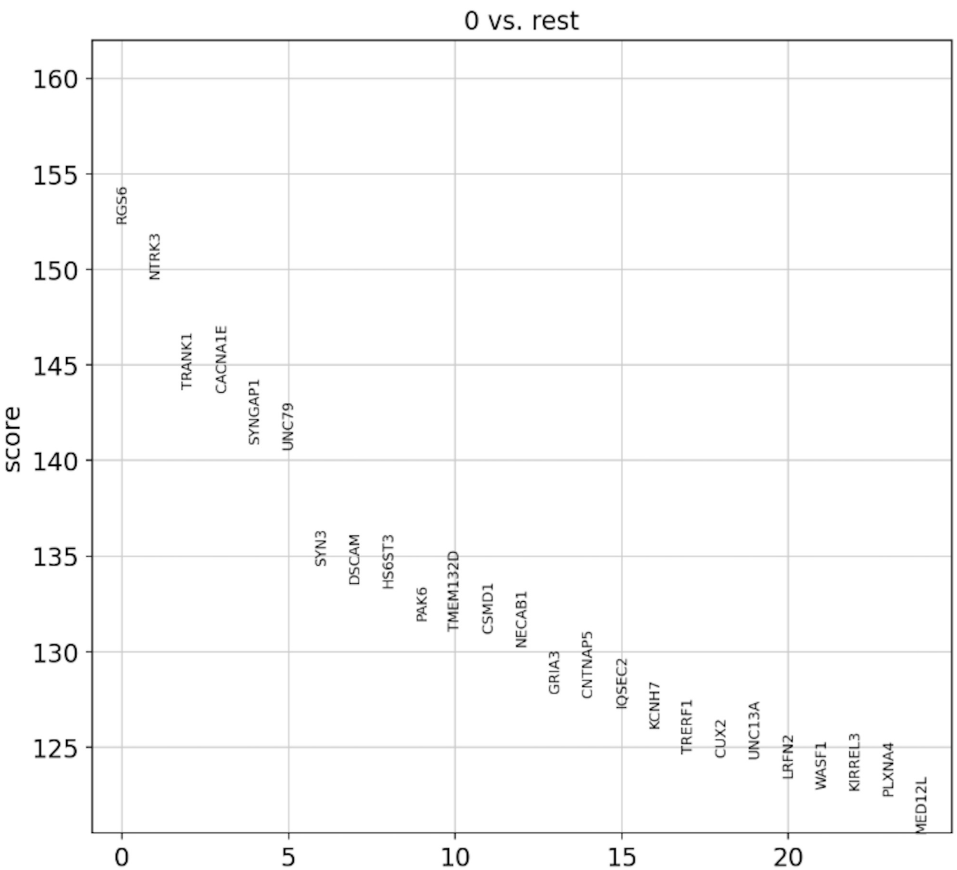


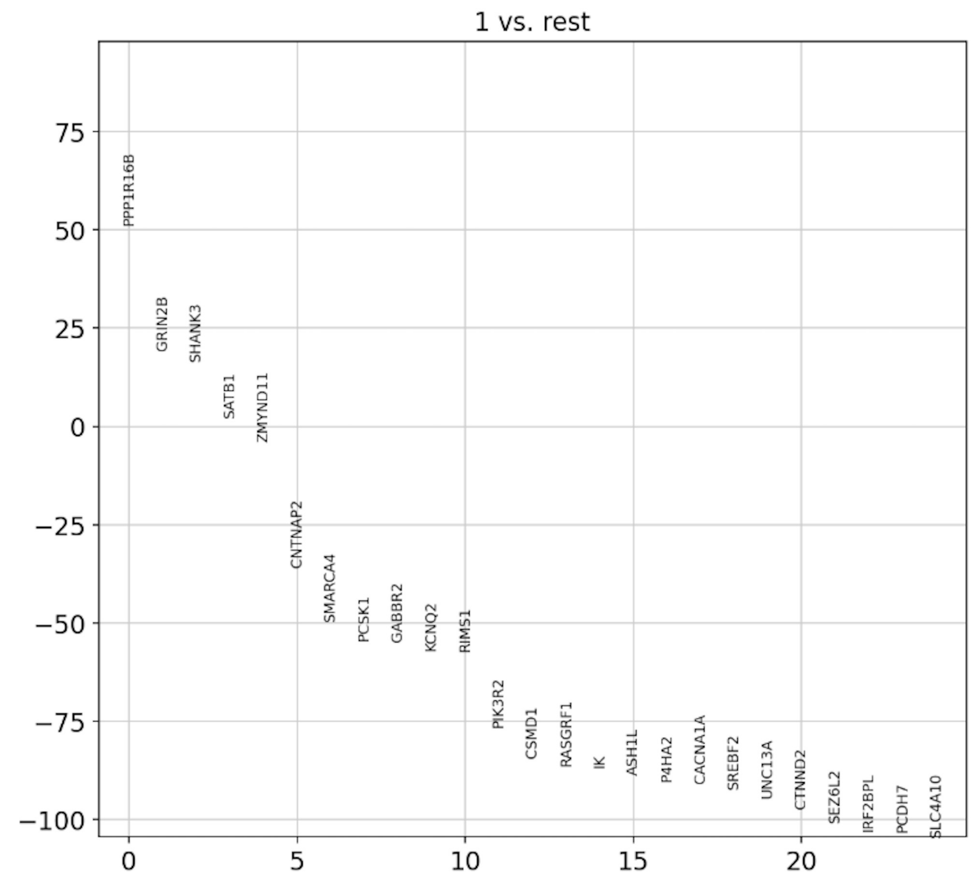
UMAP spatial clusters adata_filter



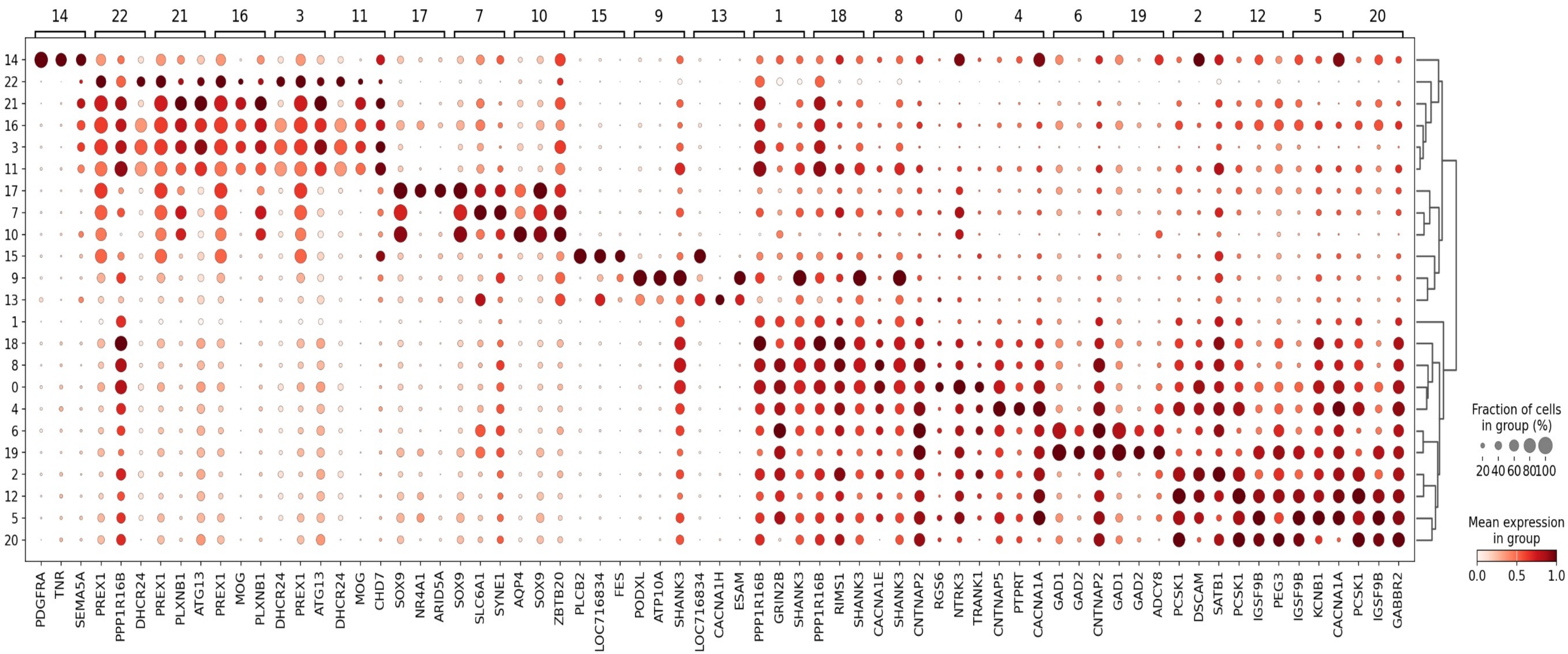


Rank Gene Groups Results

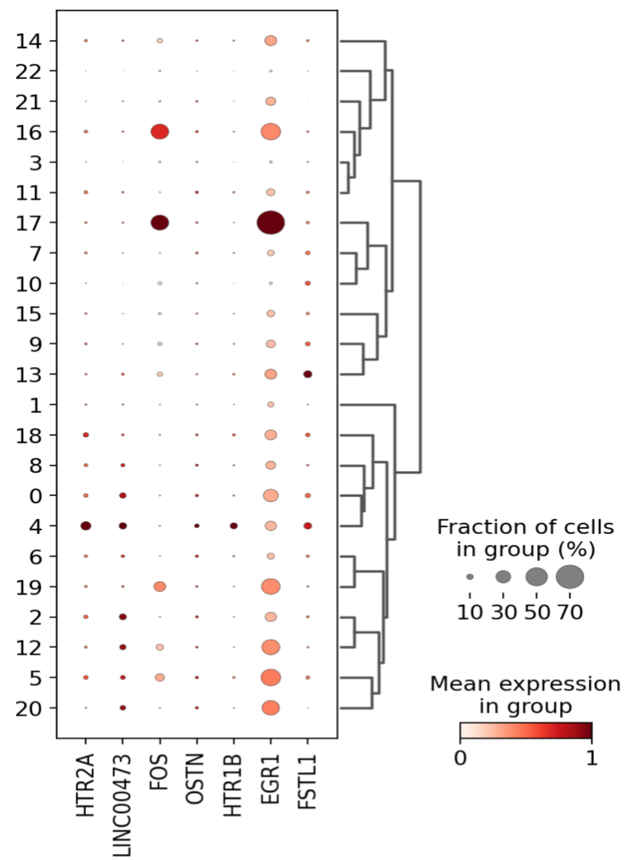




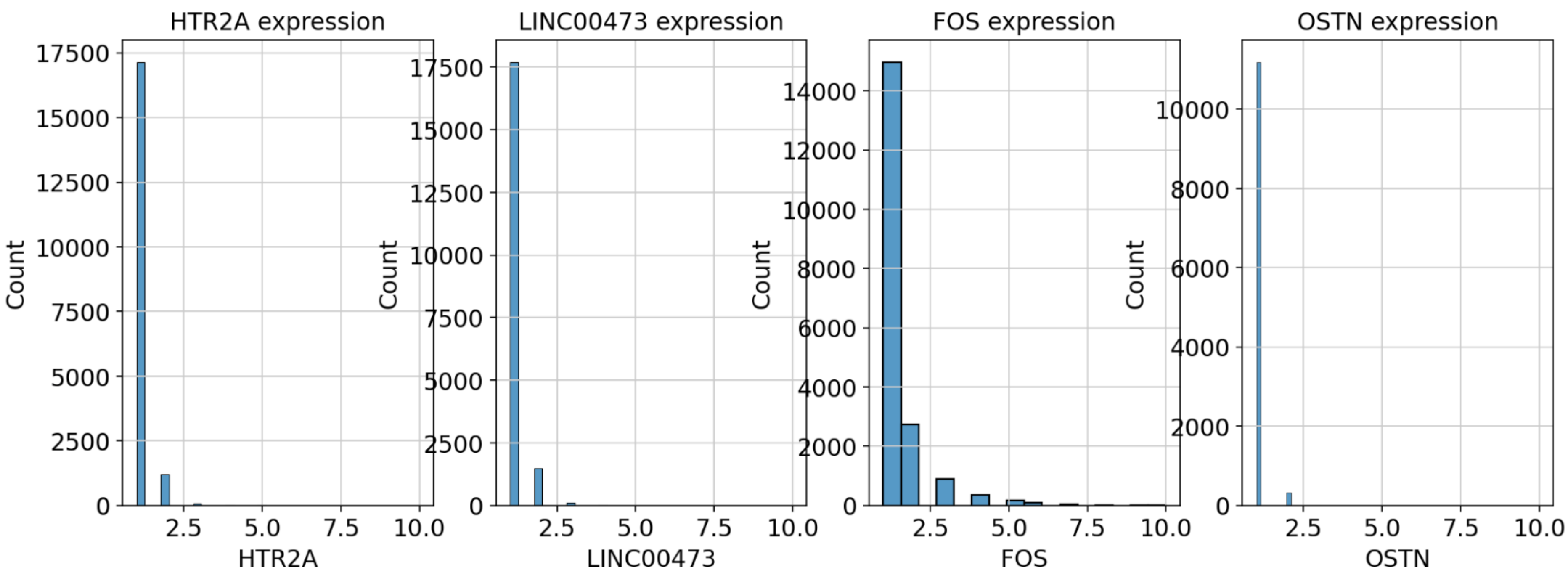
Marker Gene Expression Across Cell Clusters



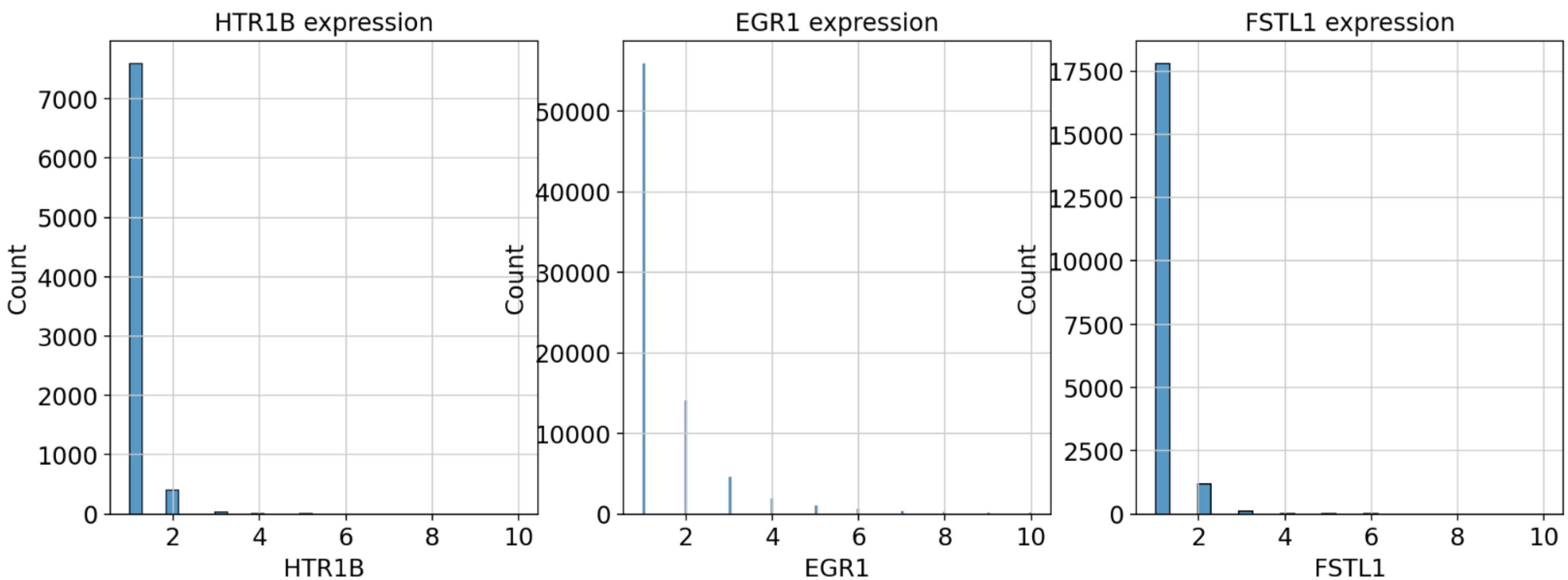
Dot Plot for 7 activity-dependent genes of interest



Transcript count per cell histograms



Transcript count per cell histograms



Identify cells with transcript count greater than 1 for 7 activity dependent genes of interest

- Out of total cells in adata_filter (348,460) how many are positive for the 7 activity-dependent genes?
- Calculate in python and export each region as csv to be read into Vizualizer

```
adata_filter.obs['HTR2A'].value_counts()
```

```
HTR2A
HTR2A_positive    16625
Name: count, dtype: int64
```

```
adata_filter
```

```
AnnData object with n_obs × n_vars = 348460 × 881
    obs: 'fov', 'volume', 'min_x', 'min_y', 'max_x', 'ma
_raw', 'MEF2C_high_pass', 'PolyT_raw', 'PolyT_high_pass'
API_high_pass', 'n_genes_by_counts', 'log1p_n_genes_by_c
ounts_in_top_100_genes', 'pct_counts_in_top_200_genes',
unts_mt', 'n_counts', 'volume_factor', 'leiden', 'manual
    var: 'mt', 'n_cells_by_counts', 'mean_counts', 'log1
cells', 'mean', 'std'
    uns: 'spatial', 'log1p', 'pca', 'neighbors', 'umap',
t_colors'
    obsm: 'blank_genes', 'spatial', 'X_pca', 'X_umap'
    varm: 'PCs'
    obsp: 'distances', 'connectivities'
```

```
# out of 348,640 cells in adata_filter object, 16,625 of
```

```
# do for rest of genes below!
```

```
adata_filter.obs['LINC00473+'].value_counts()
```

```
LINC00473+
LINC00473_positive    17323
Name: count, dtype: int64
```

```
adata_filter.obs['FOS+'].value_counts()
```

```
FOS+
FOS_positive    17520
Name: count, dtype: int64
```

```
adata_filter.obs['OSTN+'].value_counts()
```

```
OSTN+
OSTN_positive    10251
Name: count, dtype: int64
```

```
adata_filter.obs['HTR1B'].value_counts()
```

```
HTR1B
HTR1B_positive    7276
Name: count, dtype: int64
```

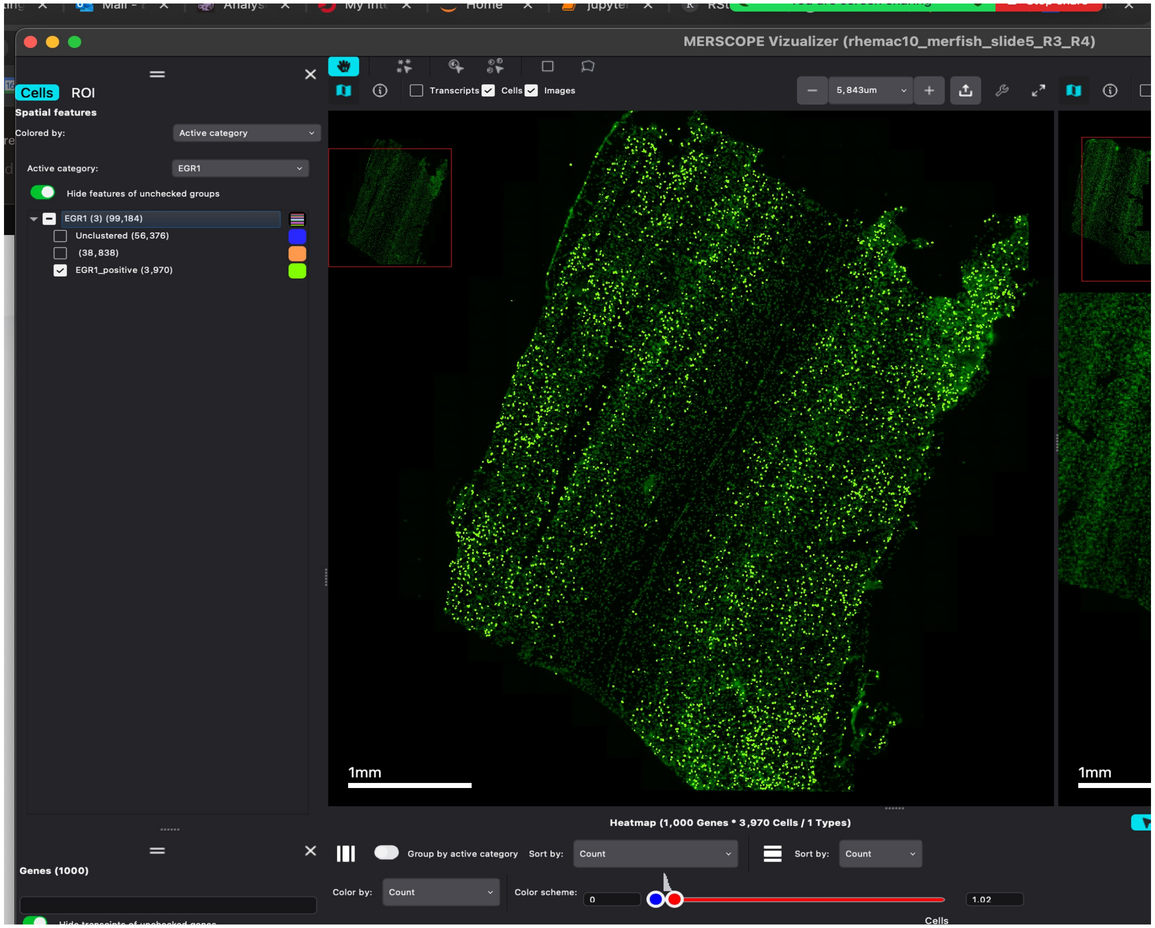
```
adata_filter.obs['EGR1+'].value_counts()
```

```
EGR1+
EGR1_positive    48687
Name: count, dtype: int64
```

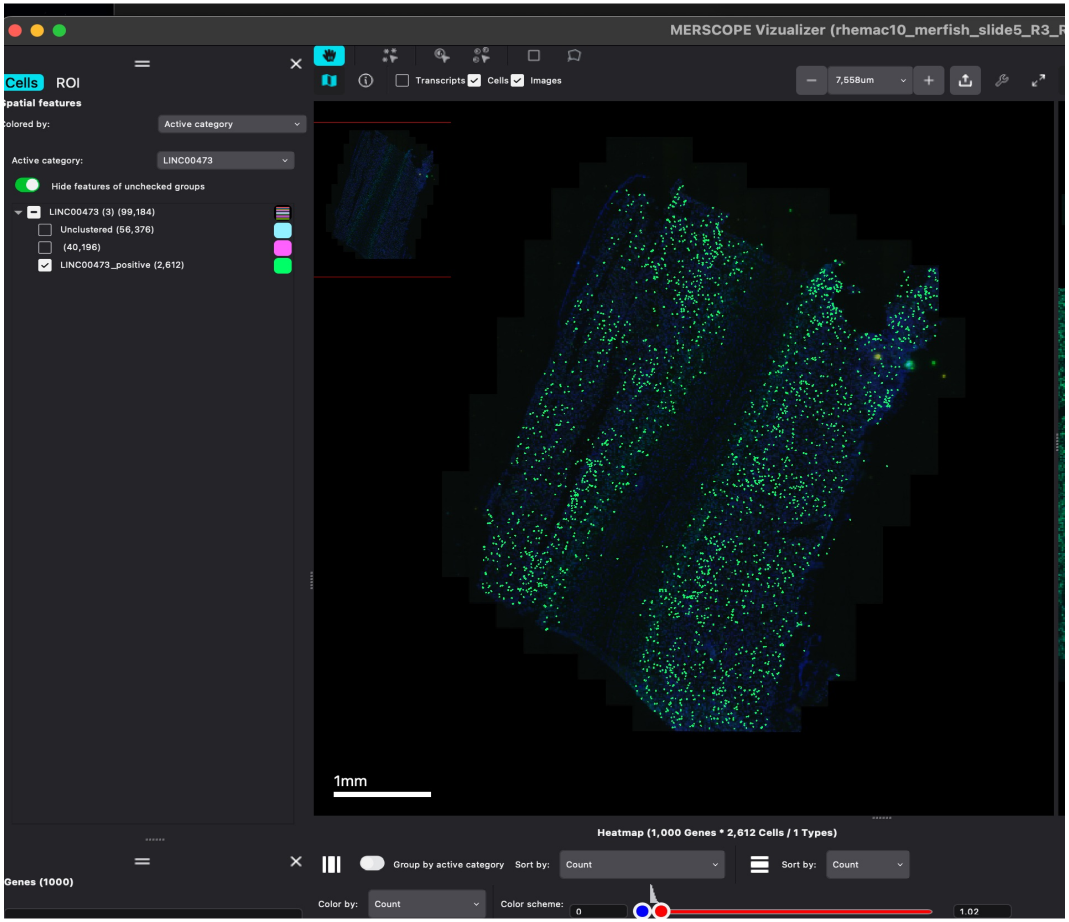
```
adata_filter.obs['FSTL1'].value_counts()
```

```
FSTL1
FSTL1_positive    17192
```

Example: Slide5 R3 EGR1



Example: Slide5 R3 LINC00473



Identify cells with transcript count greater than 2 for 7 activity dependent genes of interest

```
HTR2A_gt2
HTR2A_positive    14141
Name: count, dtype: int64
```

```
adata_filter.obs['LINC00473+_gt2'].value_counts()
```

```
LINC00473+_gt2
LINC00473_positive    14692
Name: count, dtype: int64
```

```
adata_filter.obs['FOS+_gt2'].value_counts()
```

```
FOS+_gt2
FOS_positive    14353
Name: count, dtype: int64
```

```
adata_filter.obs['OSTN+_gt2'].value_counts()
```

```
OSTN+_gt2
OSTN_positive    9788
Name: count, dtype: int64
```

```
adata_filter.obs['HTR1B_gt2'].value_counts()
```

```
HTR1B_gt2
HTR1B_positive    7187
Name: count, dtype: int64
```

```
adata_filter.obs['EGR1+_gt2'].value_counts()
```

```
EGR1+_gt2
EGR1_positive    25186
Name: count, dtype: int64
```

```
adata_filter.obs['FSTL1_gt2'].value_counts()
```

```
FSTL1_gt2
FSTL1_positive    14447
Name: count, dtype: int64
```

Identify cells with transcript count greater than 3 for 7 activity dependent genes of interest

```
HTR2A_gt3
HTR2A_positive    9927
Name: count, dtype: int64
```

```
adata_filter.obs['LINC00473+_gt3'].value_counts()
```

```
LINC00473+_gt3
LINC00473_positive    10080
Name: count, dtype: int64
```

```
adata_filter.obs['FOS+_gt3'].value_counts()
```

```
FOS+_gt3
FOS_positive    10433
Name: count, dtype: int64
```

```
adata_filter.obs['OSTN+_gt3'].value_counts()
```

```
OSTN+_gt3
OSTN_positive    7787
Name: count, dtype: int64
```

```
adata_filter.obs['HTR1B_gt3'].value_counts()
```

```
HTR1B_gt3
HTR1B_positive    6327
Name: count, dtype: int64
```

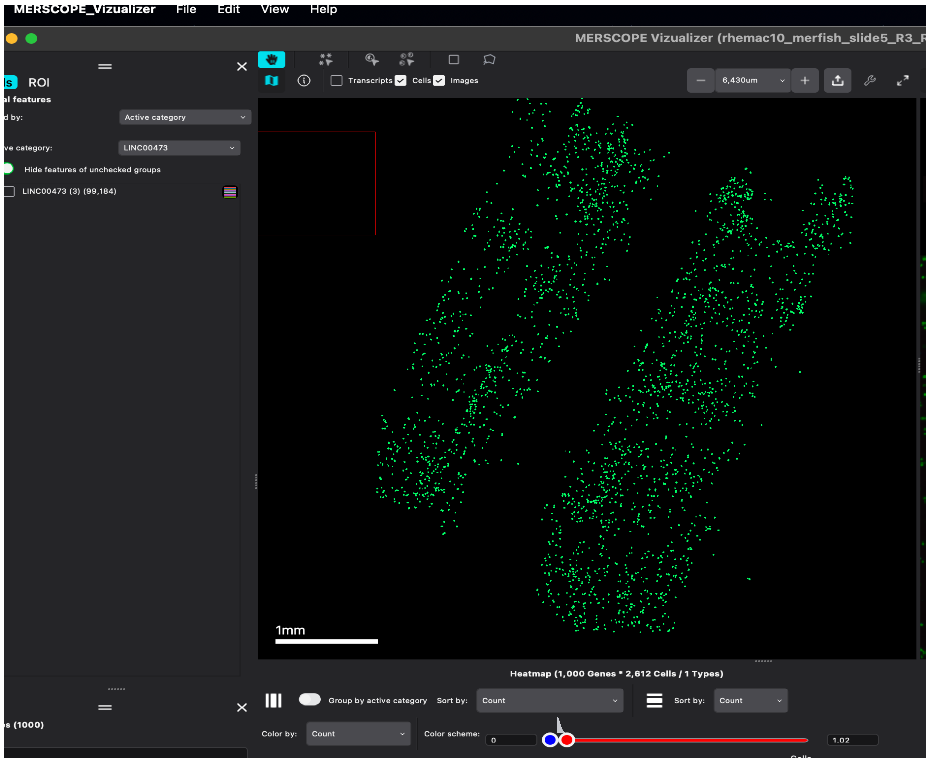
```
adata_filter.obs['EGR1+_gt3'].value_counts()
```

```
EGR1+_gt3
EGR1_positive    14585
Name: count, dtype: int64
```

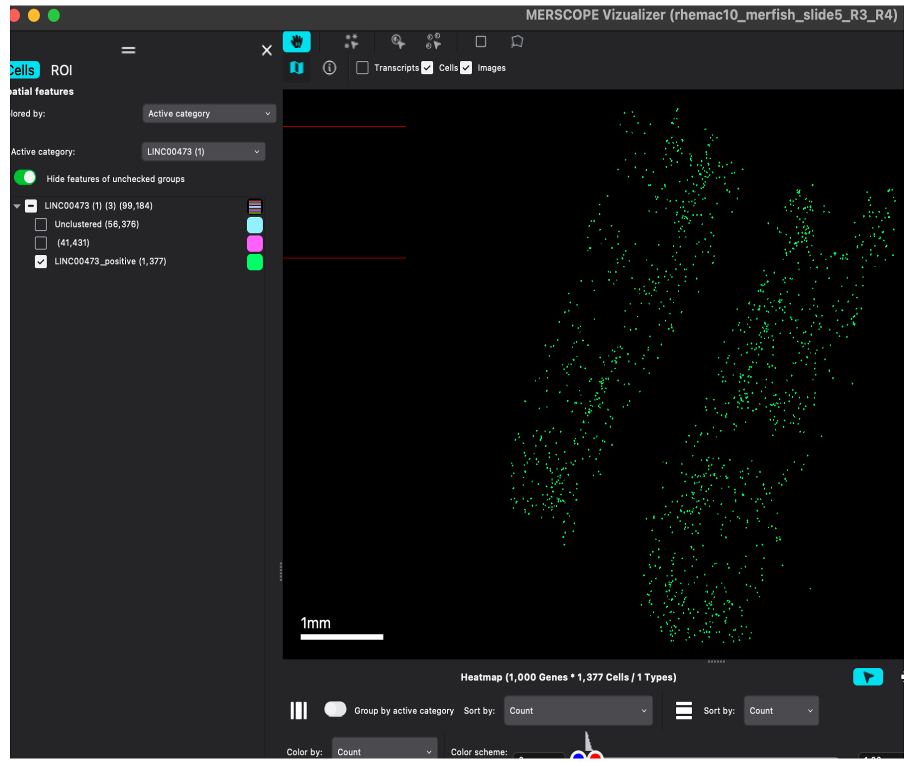
```
adata_filter.obs['FSTL1_gt3'].value_counts()
```

```
FSTL1_gt3
FSTL1_positive    10024
Name: count, dtype: int64
```

Example Slide5 R3 LINC0073 transcript > 2 vs transcript > 3



Transcript > 2



Transcript > 3

In progress/next steps

- Identify cell types in clusters based on marker genes
- Identify spatial organization of cell types
- Retrieve slide4 python objects
- Identify ODCs in rest of tissue regions in slide5 and slide4
- Collab with The Broad Institute
  - Downsampling using hexgrid to check if ODCs exist
  - Create alpha shape on positive cells to identify column regions